

Statistical consulting

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Summary The paper and the special issue focus on the activity of statistical consulting and its varieties. This includes academic consulting, consulting to and in industry as well as statistics in public media.

Keywords Statistical consulting

1 Introduction

The term ‘statistical consulting’ is nowadays widely used in the statistical community, but its meaning and the related activities are less clear and somewhat cloudy. To shed more light on this scientific field the two authors organized a workshop on statistical consulting which was held at and financed by the Center for Interdisciplinary Research (ZiF) at the University of Bielefeld in September 2006. During this workshop statisticians and applied scientists exchanged their ideas and understanding of statistical consulting. Some of the statements given during the workshop are reiterated in the following papers. The participants also discussed and disputed questions related to statistical consulting. An essence of these discussions is given below. All in all it shows that the field of statistical consulting is active and has a wide range of activities.

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2 What is statistical consulting?

For a practitioner, the process of statistical consulting resembles visiting the doctor (GP) or lawyer; it is necessary, but not necessarily popular. However, the practitioner as the client needs the insight that he or she needs help and support, just like when he or she visits the GP. This means that clients approaching the statistical consultant are aware of their needs. But the question then becomes how to get hold of those not contacting the consultant, those who are not aware that they might benefit from statistical help. Following the metaphor of the GP further, it must be clear that the consultant should act professionally and reliably, just like a GP, which applies in particular to a client that comes from outside academia. Even though there seems to be a consensus that statistical consulting should be well-structured and target-orientated, the range of activity and the process itself seems to be less well-understood. What distinguishes statistical consulting from (industry) consulting, what makes it different, what makes it special and, in general, is the word ‘statistical’ a good selling point? This brings the discussion to a general level and the question whether (modern) statistics with its features and possibilities is appropriately understood by non-statisticians. Even though the marketing aspect of statistics as a discipline is important and the marketing strategies applied by our discipline are not necessarily optimal, there have been successful attempts to make statistics more attractive. Statistical consulting itself might then be a helpful and useful instrument to change people’s view about statistics. It is therefore essential to focus this field and to promote it adequately. Along the same line we may question whether students are being taught the professionally required syllabus. The curriculum on statistics focuses more on theory than on practice, and it is often omitted to emphasize the impact of statistics and what can be done with statistics. No doubt, it is a balancing act and neither theory nor practice should rule out the other in statistical education. In fact, experiences in practical data analysis as well as insight in statistical theory are both required in statistical consulting, particularly for statisticians working in industry.

For a productive consulting project it is essential to bring potential clients as early as possible together with statisticians, which can then contribute to and accompany a project for a sufficiently long time. This then can ‘sell’ statistical knowledge and tools at various levels. It seems needless to stress that selling statistics also indicates a price, that is, statistical consulting should usually not be offered for free, particularly if projects are long term, or provided to or within industry. This brings us back to the metaphor we started with. Who considers a GP or lawyer as reliable and serious if consultation is free of charge? One clearly accepts that the GP or lawyer needs compensation. This attitude should apply to the statistician as well.

3 Statistical consulting and interdisciplinary research and collaboration

Statistical consulting can be seen as a starting point for interdisciplinary research. In fact, the seminal breakthroughs in statistics were stimulated by subject matter problems. Statistical tools and theory were developed to allow answers to data-analytic questions (see for instance Yates 1951; Kruskal 1980). This original role of statis-

tics is nowadays sometimes forgotten, since most of the statistical tools have achieved a high level of sophistication. Certainly, statistics has developed rapidly in the last three decades with computer and other technology progressing as well. The field of statistics is nowadays extremely wide with smooth transitions to neighboring and applied scientific fields. Today, it is nearly impossible to overlook the possibilities and applicability of statistics in its full range. On the other hand, traditional problems remain as important and must be focused on, as they were decades ago. For instance, statistics is used to support political decisions, and forecasting is an essential business for economies. But expressing and comprehending uncertainty, e.g., in terms of prediction intervals, is still a delicate and awkward issue for numerous politicians and other practitioners. This does not mean that statistical consulting should concentrate on essentials (from a statistician's point of view) only, but basic concepts and ideas might be less understood by non-statisticians than assumed by statisticians. In other words, digging deeper and deeper into statistical methodology does not necessarily promote the discipline if traditional problems are not communicated as they should be. Practitioners want simple solutions and these need to be only as sophisticated as necessary. Therefore, to fully promote statistical consulting and to stimulate interdisciplinary research the consultant should use his or her full range of intellectual capacity and should contribute to all levels, possibly with simple approximate methods if these solve the applied problems sufficiently. But the question remains how statistics can promote research and interdisciplinary collaboration. In fact, the problem is not related to the clients approaching a statistician or a statistical consultancy unit requesting support. The problem occurs with those scientists who are not aware of the fact that a statistical expert could help. This raises again the point that statistics needs promotion in order to be seen by other scientists and practitioners. Such promotion can be started within a university by offering courses to non-statisticians, by building up contact to those scientists who make use of or collect data. On the other hand, if clients are approaching the statistician it is important to keep track of them and to support them. Criticizing the clients' work is unproductive and a bad starting and selling point. It is more important to build up faith and explain and teach a better and more sound statistical analysis. In fact, appropriate teaching and consulting is establishing connections, and such connections can lead to interdisciplinary research, but at least to mutual benefits.

4 Selling statistics, making people read

Statisticians have the nickname 'number crunchers' and statistics is seldom seen in the public media in a way that a statistician would feel pleased. Journalists and statisticians are like cat and mouse and what is printed as statistics in journals and books is often far from the discipline a statistician calls his or her own. For instance, ranking became popular in the last years with, e.g., university rankings being published every other week. There is, no doubt, a statistical core in ranking and, in fact, variability is essential but often omitted entirely. And even though the statistical tool-box has models and methods to assess results based on rankings scientifically, statistics

is seldom (or never) applied to results from a ranking process. This is just one example of how statistics is used or not used in public media. Although selling statistics or statistical stories to a wide public is a difficult task, this applies in the same way to most sciences. A journalist has the job to sell readable and fascinating stories, so his or her first question has to be: ‘What is the story?’ The scientist on the other hand feels tempted by the methodology or discoveries behind the story and wants these to be told. A hidden conflict is therefore inevitable. Nevertheless, journalists need scientists and scientists need journalists. It is again a compromise and a challenge for both sides to interact, and this interaction is part of statistical consulting as well. As scientists, however, we have to observe the rules of the journalism game as these are different from those in the scientific world. For journalists time is much more an issue than for scientists, newspaper articles have to be written now, not tomorrow or next week. The article should have a story, not formulas or derivations. And, a new story for a wide readership and public might be old or boring for the scientist working in the field, but again, the journalists know the rules of the game, not the scientists. Apparently, there is no obligation for the scientist to collaborate with or support a journalist, but the scientist can help the journalist to promote statistics on a wider level simply by reporting statistical results soundly. In this respect the issue of trust is essential. Scientists and journalists have to break down barriers and may collaborate, but this needs both time and faith.

5 Special Issue on statistical consulting

The subsequent papers collected in this Special Issue give personal views of statistical consulting, either in general or with a special focus. Antony Unwin as well as Adrian Bowman discuss the aspects of interaction. The authors stress the importance of communication skills to achieve fruitful collaboration. Windmann and Kauermann report the results of a survey which mirrors the current state of the art of statistical consulting at German universities. The paper by Christoph Schmidt shows how statistical consulting is seen and used in economic research and for political advice. The role of statistics in the media is described by Holger Wormer from the journalists’ perspective. The remaining papers concentrate on statistical consulting in industry. Christoph Berding and Wilhelm Kleider give an example from the pharmaceutical industry and illuminate the role of statistics in this field. A more generic view is discussed in Christian Ritter’s contribution, reporting on the role of a statistical consultant in a special industry project. Along the same line is the paper by Jan Engel and Henriette Hoonhout, mirroring experiences of statistical consultants in industry. Finally, Claus Weihs concludes with attempts to measure and control the quality of statistical consulting itself.

6 Summary

The above remarks and the subsequent papers mirror the various aspects of statistical consulting and its widespread activities. We think that the contributions show that the field of statistical consulting is dynamic and worthwhile.

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